

# tmux documentation

## Working Plan — v1.0.4 cycle (2026-05-21)

**Revision:** 1 **Last modified:** 2026-05-21T00:00:00Z **Scope:** anti-bluff covenant propagation + CodeGraph (§11.4.78) + audit follow-ups

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### §1 Triggering directives

1. **User mandate (2026-05-21):** "create full working plan for tackling all these points completely ... all existing tests and Challenges MUST work in anti-bluff manner ... must guarantee the quality, the completion and full usability by end users of the product! This MUST BE part of Constitution ... its CLAUDE.MD and AGENTS.MD if it is not there already, and to be applied to all Submodules's Constitution, CLAUDE.MD and AGENTS.MD as well ... Pay attention that we now use and incorporate fully the HelixConstitution submodule responsible for root definitions."
2. **User mandate (2026-05-21, follow-up):** "Incorporate / install codegraph into the our project ... installed for Claude Code, OpenCode, Kimi CLI, Crush, Qwen Code ... create comprehensive tests to validate and verify with anti-bluff approach that codegraph is working completely as expected!"
3. **My honest audit (2026-05-21):** M4/M5 paired-mutations use raw GNU sed -i (silently SKIP on Darwin); tmx kill shorthand in README/AGENTS table doesn't resolve (must say kill-session).

### §2 Constitutional authority

- §11.4 end-user-quality covenant (verbatim user mandate 2026-04-28)
- §11.4.2 recorded-evidence requirement
- §11.4.4 / §103 four-layer test coverage
- §11.4.10 credentials-handling mandate (codegraph secret exclusions)
- §11.4.20 / §11.4.70 subagent-driven by default
- §11.4.26 constitution-submodule update workflow (fetch first — done)
- §11.4.30 .gitignore + no-versioned-build-artifacts (codegraph DB)
- §11.4.32 post-constitution-pull validation sweep
- §11.4.65 universal Markdown export (HTML+PDF siblings)
- §11.4.71 pre-push fetch + investigate + integrate
- §11.4.74 submodule-catalogue-first discovery
- §11.4.77 regeneration-mechanism-required (codegraph DB)
- §11.4.78 CodeGraph code-intelligence mandate
- §102 / §11.4.7 operator-path test coverage
- §12.10 continuation-document maintenance

## §3 Audit state (captured this session)

### §3.1 Verbatim user-mandate (2026-04-28) presence

| File                                                          | Hits    | Decision                                                                          |
|---------------------------------------------------------------|---------|-----------------------------------------------------------------------------------|
| Project Constitution.md                                       | 1       | ✓ keep                                                                            |
| Project CLAUDE.md                                             | 0       | <b>insert verbatim block</b>                                                      |
| Project AGENTS.md                                             | 0       | <b>insert verbatim block</b>                                                      |
| Project QWEN.md                                               | 0       | <b>insert verbatim block</b>                                                      |
| constitution/Constitution.md                                  | 1       | ✓ upstream                                                                        |
| constitution/CLAUDE.md                                        | 1       | ✓ upstream                                                                        |
| constitution/AGENTS.md                                        | 1       | ✓ upstream                                                                        |
| constitution/QWEN.md                                          | 0       | <b>upstream gap — separate PR to HelixConstitution out of scope here; note it</b> |
| Containers/CONSTITUTION.md<br>(same inode as Constitution.md) | 2       | ✓                                                                                 |
| Containers/CLAUDE.md                                          | 3       | ✓                                                                                 |
| Containers/AGENTS.md                                          | 3       | ✓                                                                                 |
| Containers/QWEN.md                                            | missing | <b>out-of-scope — Containers is owned but separate cycle</b>                      |

The four ×-gap rows in the project layer are the work for this cycle. The upstream constitution/QWEN.md gap and the Containers/QWEN.md gap are honest-noted but deferred to their own commit cycles (different submodule ownership boundaries, per §11.4.26 step 4 — those need their own fetch+pull+commit+push in their own repos).

### §3.2 CodeGraph (§11.4.78) state

- CodeGraph CLI **already installed** (/opt/homebrew/opt/node@25/bin/codegraph, v0.6.8).
- .codegraph/ not yet present in the repo — needs codegraph init + codegraph index.
- MCP wiring state per agent:

| Agent                        | Config path                           | Exists? | Needs                       |
|------------------------------|---------------------------------------|---------|-----------------------------|
| Claude Code (project-scoped) | .mcp.json                             | ×       | create with codegraph entry |
| OpenCode                     | ~/ .config/opencode/<br>opencode.json | ✓       | add codegraph entry         |
| Kimi CLI                     | ~/ .kimi/mcp.json                     | ✓       | add codegraph entry         |
| Crush                        | .crush.json                           | ×       | create with codegraph entry |
| Qwen Code                    | .qwen/settings.json                   | ×       | create with codegraph entry |

### §3.3 Audit follow-ups (from "what's left unfinished")

| ID      | Issue                                                            | Severity                 |
|---------|------------------------------------------------------------------|--------------------------|
| AUDIT-1 | M4/M5 use raw <code>sed -i</code> (GNU-only) → SKIP on Darwin    | Minor (gap, not failure) |
| AUDIT-2 | <code>tmx kill -t NAME</code> ambiguous (README claims it works) | Minor doc/UX bug         |
| AUDIT-3 | CONTINUATION.md §4 self-referential placeholder SHA stale        | Cosmetic                 |

## §4 Phased implementation

Each phase ends with a captured-evidence checkpoint per §11.4.2; each non-trivial code path lands all four §103 layers (static gate / runtime test / challenge / paired mutation).

### Phase 1 — Anti-bluff covenant propagation (project layer)

- **1.1** Insert verbatim user-mandate (2026-04-28) + §11.4 covenant summary into project `CLAUDE.md` (in a clearly-labeled `## MANDATORY ANTI-BLUFF COVENANT` (verbatim from upstream §11.4) section, BELOW the existing inheritance pointer so it doesn't break the `@import` semantics).
- **1.2** Same into project `AGENTS.md`.
- **1.3** Same into project `QWEN.md`.
- **1.4** Add `scripts/tests/19_anti_bluff_covenant_propagation.sh` (operator-path: greps each governance file for the literal quote; FAILs if any file lacks it).
- **1.5** Static gate in `verify.sh`: pre-suite check that all four governance files carry the literal anchor `We had been in position that all tests do execute with success.`
- **1.6** Challenge `TMUX-CH-19` in `scripts/challenges/tmux.yaml`.
- **1.7** Paired mutation M15: strip the literal from a TEMP COPY of `CLAUDE.md`, run test 19, assert FAIL; restore, assert PASS.
- **Evidence:** test 19 + `verify.sh layer-1` + meta-test M15.

### Phase 2 — CodeGraph install + initialisation (§11.4.78)

- **2.1** `codegraph init` from repo root — produces `.codegraph/config.json`.
- **2.2** Edit `.codegraph/config.json` `exclude` list to add (per §11.4.10):
  - `constitution/` (other-owned submodule)
  - `Containers/` (other-owned submodule)
  - `tmux/` (upstream third-party submodule)
  - `.env*`, `*.env`, `*.pem`, `*.key`, `*.crt`, `id_rsa*`, `id_ed25519*`, `secrets/`, `.netrc`
  - `tmux/build*/`, `out/`, `.codegraph/codegraph.db*`
- **2.3** Add `.codegraph/codegraph.db` (and `.codegraph/codegraph.db-*`) to project `.gitignore`. `.codegraph/config.json` stays tracked.
- **2.4** Document the regeneration mechanism per §11.4.77 at `.gitignore-meta/codegraph-db.yaml`: `pattern`, `mechanism-type=regenerate`, `script-path=scripts/codegraph_reindex.sh`, `expected-disk-usage`, `vendor-url`, `requires-network=false`, `requires-credentials=false`.

- **2.5** `scripts/codegraph_reindex.sh` — idempotent script that runs `codegraph index` and writes the stamp file `.gitignore-meta/.regenerated/codegraph-db.ok`.
- **2.6** Wire into `scripts/setup.sh` as a post-build step so a fresh clone gets a usable index.
- **Evidence:** `.codegraph/codegraph.db` materialised, node count non-zero captured via `codegraph status`.

## Phase 3 — Multi-agent MCP wiring

For each of the 5 agents listed in §11.4.78:

- **3.1 Claude Code** — create project-scoped `.mcp.json`:

```
{
  "mcpServers": {
    "codegraph": {
      "type": "stdio",
      "command": "codegraph",
      "args": ["serve", "--mcp"],
      "env": {}
    }
  }
}
```

- **3.2 OpenCode** — read `~/.config/opencode/opencode.json`, merge a `codegraph` entry under `mcp.servers`, write back atomically.
- **3.3 Kimi CLI** — read `~/.kimi/mcp.json`, merge a `codegraph` entry, write back atomically.
- **3.4 Crush** — create `.crush.json` with the `codegraph` MCP server.
- **3.5 Qwen Code** — create `.qwen/settings.json` with the `codegraph` MCP server.
- Every config references the bare `codegraph` command on `PATH`; no hardcoded host paths (so configs are portable across machines).

## Phase 4 — Anti-bluff CodeGraph verification suite (§11.4.78)

The mandate **REQUIRES** an unforgeable challenge — a fact obtainable only by calling a CodeGraph MCP tool, so an agent answering from file-reading cannot produce a false **PASS**.

- **4.1** `scripts/tests/20_codegraph_installed.sh` — verifies `codegraph` on `PATH`, version  $\geq 0.6.0$ , `.codegraph/config.json` present, secret exclusions present.
- **4.2** `scripts/tests/21_codegraph_index_present.sh` — verifies `.codegraph/codegraph.db` exists, runs `codegraph status` (or equivalent CLI), captures node count  $> 0$  as positive runtime evidence.
- **4.3** `scripts/tests/22_codegraph_mcp_wired.sh` — verifies each of the 5 agent configs references `codegraph` MCP server. Operator-path: parses each config JSON; SKIPS the config-path check for agents whose binary isn't installed on this host (and records **WHY** per §11.4.3).
- **4.4** Unforgeable-challenge per §11.4.78: end-to-end test that drives Claude Code (or the available CLI agent) with a prompt asking for a CodeGraph-only fact. Per §11.4.3 if no agent CLI is available on this host the test SKIPS with documented reason (**AUTONOMOUS\_DEIGNED** per §11.4.52 — falls in the "agent that genuinely cannot be driven end-to-end" carve-out the mandate enumerates).

- **4.5** Challenges TMUX-CH-20, TMUX-CH-21, TMUX-CH-22 in yaml.
- **4.6** Paired mutations (3): M16 strips secret exclusion from `config.json`; M17 removes codegraph entry from `.mcp.json`; M18 deletes `.codegraph/codegraph.db` stamp.
- **Evidence:** every test runs, every mutation caught.

## Phase 5 — AUDIT-1: M4/M5 portability fix

- **5.1** Replace inline `sed -i 's|...|...' "$target_abs"` in M4/M5 with a call to the portable `inplace_sed` helper already added in A17.
- **5.2** Re-run meta-test on Darwin; expect SKIP count 6 → 4 (M7-M10 remain SKIP — they need `scripts/tmx-vm` which is the Linux-in-VM wrapper, not present on Darwin host — that's correct §11.4.3 topology dispatch).
- **Evidence:** before/after meta-test transcript showing SKIP count drop.

## Phase 6 — AUDIT-2: `tmx kill` shorthand fix

- **6.1** Wrapper: detect `tmx kill <args>` (bare verb) and translate to `tmx kill-session <args>`. Don't break `tmx kill-pane / kill-server / kill-window` — only intercept the bare unambiguous `kill` verb.
- **6.2** Update README + AGENTS commands table — still lists `tmx kill` as the friendly verb, now actually works.
- **6.3** Test 23 `scripts/tests/23_tmx_kill_shorthand.sh` — creates a session, kills via `tmx kill -t NAME`, asserts session removed. Operator-path per §102.
- **6.4** Paired mutation M19: remove the shorthand translation, assert test 23 FAILs.
- **Evidence:** test 23 captures session before/after.

## Phase 7 — Documentation + sync (per §11.4.65 + §12.10)

- **7.1** `docs/codegraph/README.md` — Overview / Prerequisites / Install / Init / MCP wiring per agent / Anti-bluff verification / Troubleshooting.  
◦ HTML + PDF siblings per §11.4.65.
- **7.2** `Fixed.md`: A18 (CodeGraph integration), A19 (anti-bluff covenant propagation), A20 (M4/M5 portability), A21 (`tmx kill` shorthand).
- **7.3** `CHANGELOG.md`: [v1.0.4] block.
- **7.4** `VERSION`: bump to 1.0.4 / `versionCode=5`.
- **7.5** `CONTINUATION.md` §3.10 (this cycle); §1 snapshot table refresh.
- **7.6** `Issues.md`: zero open items confirmed; record the upstream gap (constitution/QWEN.md + Containers/QWEN.md missing the covenant) as a NEW E1 entry blocking on separate-submodule cycles.

## Phase 8 — §11.4.71 pre-push + §11.4.32 post-pull sweep + commit + push

- **8.1** `git fetch --all --prune --tags parent` + every owned submodule (re-run; integration already done at start of this cycle).
- **8.2** Re-run full verify gate fresh (per §11.4 Iron Law).
- **8.3** Re-run meta-test fresh (expect 21 caught / 0 escaped / 4 skipped — 18 prior + 3 new from Phase 4 mutations, minus M4/M5 portability fix dropping their SKIP).
- **8.4** Re-run e2e test fresh.
- **8.5** `commit_all.sh "<v1.0.4 message>"` → push to github + gitlab.
- **Evidence:** full transcript triple captured this session.

## §5 Risk register

| Risk                                                                                                                         | Mitigation                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>codegraph init</code> discovers project-specific config schema → first-class fields differ from what §11.4.78 mandates | Read actual generated config first; adapt secret-exclusion list to schema                                                                               |
| MCP config formats differ per agent                                                                                          | Each agent's docs consulted before wiring; configs validated by parsing JSON post-write                                                                 |
| <code>.codegraph/codegraph.db</code> could be very large (multi-MB)                                                          | Honest size capture in <code>.gitignore-meta/codegraph-db.yaml</code> ; gitignored                                                                      |
| End-to-end MCP test requires running an agent CLI in non-interactive mode                                                    | Per §11.4.78 carve-out, document SKIP for agents that cannot be driven (with reason); the helper-presence + parse-validation gates are still mechanical |
| Editing constitution/ submodule out of scope (§11.4.26 step 1+)                                                              | Upstream gaps logged as separate work items; this cycle touches consumer side only                                                                      |

## §6 Out-of-scope (logged, not done this cycle)

- Upstream `constitution/QWEN.md` add covenant — needs separate PR to `HelixDevelopment/HelixConstitution`.
- `Containers/QWEN.md` create + populate — needs separate PR to `vasic-digital/Containers`.
- Linux-host SKIPS (M7-M10 + tests 08, 09, 12, 13, 14) — they require Linux+cgroup-v2+systemd; no Darwin path exists.
- XNU memory-cap gap — kernel-level, not addressable.

## §7 Approval gate

Plan covers four cycle goals:

1. (Mandate) anti-bluff covenant verbatim in every consumer governance file.
2. (Mandate) CodeGraph installed + wired for all 5 agents + verified anti-bluff.
3. (Audit) M4/M5 portability + `tmx kill` shorthand.
4. (Discipline) §103 four-layer coverage on every new behaviour, §11.4.65 doc sync, §11.4.71 pre-push, §12.10 CONTINUATION, commit via `commit_all.sh`.

Estimated commits: 1 (everything in one v1.0.4 cycle commit per §11.4.42 iteration discipline).  
Approve to proceed with execution.